THE DE RHAM COMPLEX OF LOG POLES ALONG AN EFFECTIVE CARTIER DIVISOR SITS IN A CANONICAL SHORT EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The comparison of a module with its twist by a Cartier divisor, giving the inclusion of the de Rham complex. More on Flatness, Section \ref{flat-section-eta} (invoked at proof line 2423). It is the reference for the canonical inclusion $\Omega^p_{X/S} \subset \Omega^p_{X/S}(Y)$ and the statement that the differentials have no nonzero local sections supported on Y.
- The shift convention for cochain complexes. Homology, definition homology-definition-shift-cochain (invoked at proof line 2466). It fixes the sign of the differential on $\Omega^\bullet_{Y/S}[-1]$, which the author matches with the forced sign in the Leibniz rule for the log complex.
- The local splitting hypothesis defining log poles. Same chapter derham.tex, the assumption stated in the statement (used at proof lines 2405-2417). For a local equation f it gives $\Omega_{X/S,y} = \mathcal{O}_{X,y}\mathrm{d}f \oplus M$ with torsion free exterior powers, from which $\Omega^p_{Y/S,y} = \wedge^p(M/fM)$ and the absence of sections supported on Y follow.

Source reference. Original section 0F8L: sources/stacks-chow/tags.txt, source lines 19046–19046. Navigation pointer for \ref{flat-section-eta} used inside the counted span (line 2423); the argument that consumes it is retained in the span.

Source reference. Original definition 011G: sources/stacks-chow/tags.txt, source lines 1362–1362. Navigation pointer for \ref{homology-definition-shift-cochain} used inside the counted span (line 2466); the argument that consumes it is retained in the span.

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Lemma 0.1. *Let $X \to S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S. There is a canonical short exact sequence of complexes*

$$0 \to \Omega^\bullet_{X/S} \to \Omega^\bullet_{X/S}(\log Y) \to \Omega^\bullet_{Y/S}[-1] \to 0$$

Proof. Our assumption is that for every $y \in Y$ and local equation $f \in \mathcal{O}_{X,y}$ of Y we have

$$\Omega_{X/S,y} = \mathcal{O}_{X,y}\mathrm{d}f \oplus M \quad \text{and} \quad \Omega^p_{X/S,y} = \wedge^{p-1}(M)\mathrm{d}f \oplus \wedge^p(M)$$

for some module M with f-torsion free exterior powers $\wedge^p(M)$. It follows that

$$\Omega^p_{Y/S,y} = \wedge^p(M/fM) = \wedge^p(M)/f \wedge^p (M)$$

Below we will tacitly use these facts. In particular the sheaves $\Omega^p_{X/S}$ have no nonzero local sections supported on Y and we have a canonical inclusion

$$\Omega^p_{X/S} \subset \Omega^p_{X/S}(Y)$$

see More on Flatness, Section 0F8L. Let $U = \mathrm{Spec}(A)$ be an affine open subscheme such that $Y \cap U = V(f)$ for some nonzerodivisor $f \in A$. Let us consider the $\mathcal{O}_U$-submodule of $\Omega^p_{X/S}(Y)|_U$ generated by $\Omega^p_{X/S}|_U$ and $\mathrm{d}\log(f) \wedge \Omega^{p-1}_{X/S}$ where $\mathrm{d}\log(f) = f^{-1}\mathrm{d}(f)$. This is independent of the choice of f as another generator of the ideal of Y on U is equal to uf for a unit $u \in A$ and we get

$$\mathrm{d}\log(uf) - \mathrm{d}\log(f) = \mathrm{d}\log(u) = u^{-1}\mathrm{d}u$$

which is a section of $\Omega_{X/S}$ over U. These local sheaves glue to give a quasi-coherent submodule

$$\Omega^p_{X/S} \subset \Omega^p_{X/S}(\log Y) \subset \Omega^p_{X/S}(Y)$$

Let us agree to think of $\Omega^p_{Y/S}$ as a quasi-coherent $\mathcal{O}_X$-module. There is a unique surjective $\mathcal{O}_X$-linear map

$$\mathrm{Res} : \Omega^p_{X/S}(\log Y) \to \Omega^{p-1}_{Y/S}$$

defined by the rule

$$\mathrm{Res}(\eta' + \mathrm{d}\log(f) \wedge \eta) = \eta|_{Y \cap U}$$

for all opens U as above and all $\eta' \in \Omega^p_{X/S}(U)$ and $\eta \in \Omega^{p-1}_{X/S}(U)$. If a form η over U restricts to zero on $Y \cap U$, then $\eta = \mathrm{d}f \wedge \eta' + f\eta''$ for some forms η' and η'' over U. We conclude that we have a short exact sequence

$$0 \to \Omega^p_{X/S} \to \Omega^p_{X/S}(\log Y) \to \Omega^{p-1}_{Y/S} \to 0$$

for all p. We still have to define the differentials $\Omega^p_{X/S}(\log Y) \to \Omega^{p+1}_{X/S}(\log Y)$. On the subsheaf $\Omega^p_{X/S}$ we use the differential of the de Rham complex of X over S. Finally, we define $\mathrm{d}(\mathrm{d}\log(f) \wedge \eta) = -\mathrm{d}\log(f) \wedge \mathrm{d}\eta$. The sign is forced on us by the Leibniz rule (on $\Omega^\bullet_{X/S}$) and it is compatible with the differential on $\Omega^\bullet_{Y/S}[-1]$ which is after all $-\mathrm{d}_{Y/S}$ by our sign convention in Homology, Definition 011G. In this way we obtain a short exact sequence of complexes as stated in the lemma. □